

ANSWER

1. In a big data context, why data is called big data?

In context of big data the data is called the big data because of the following reasons;

- **Volume** :The vast amount of data collected from various sources
- **Velocity**: The rate at which data is generate
- **Variety**: Means the data are collected into different format
- **Veracity**: Means the accuracy and meaningfull data which can differ between the sources.
- **Value**: means the potential business value derived from the data

2. Big data is being characterized by 5Vs: Read the following statements, then from each statement attach the corresponding V.

- The vast amounts of data that are collected from multiple heterogeneous sources. **volume**
- The various formats of structured, semi-structured and unstructured data, from social media, IoT devices, database tables, web applications, streaming services, machinery, business software and other sources. **variety**
- The potential business value of the collected data. **value**
- The ever-increasing rate at which data is being generated on all fronts in all industries. **velocity**
- The degree of accuracy of collected data, which can vary significantly from one source to the next. **Veracity**

3. In big data, data can be classified as structured, semi-structured, and unstructured data. Briefly explain why data in a data lake can not be taken as structured data?

ANS: because the data lakes is design to store the raw data in a its original format which include structured data, semi structured data and unstructured data. But the structured data consist of defined format or schema , however in data lakes most of data are in undefined schema thus making it inappropriate to classify all data in the data lakes as a structured.

4. Do you think data lake can be categorized as semi-structured or unstructured data. Explain your answer.

ANS: The data lakes can be categorized as both semi structured and unstructured data this is because the data lakes store the raw data in its various format and most of its lacks predefined structure